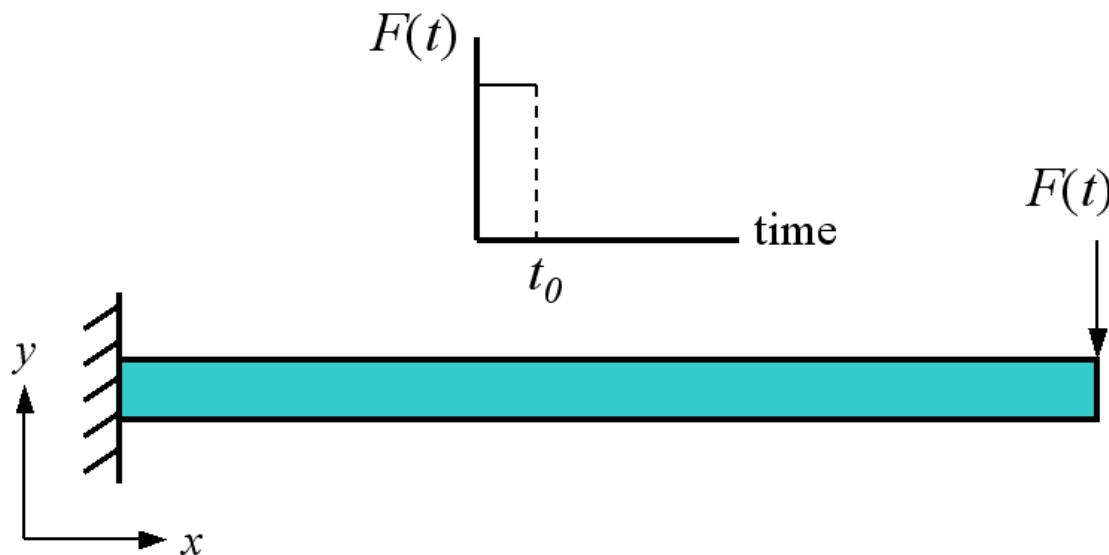
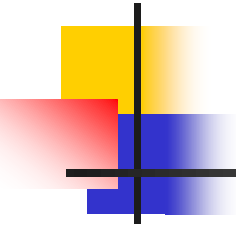


暫態動力學分析

- ❖ 暫態動力學(**transient dynamic**)分析是關於時間和加速度相關的應力與變形分析。典型的暫態動力學分析如下圖所示，當一結構受到衝擊力(**impact force**) $F(t)$ 作用時，它的反應會與時間有關，這與靜態分析大不相同，而且衝擊力作用時間 t_0 的長短也會造成不同的結構反應與應力場。



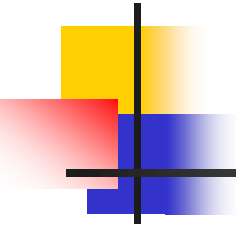


Equation of Motion

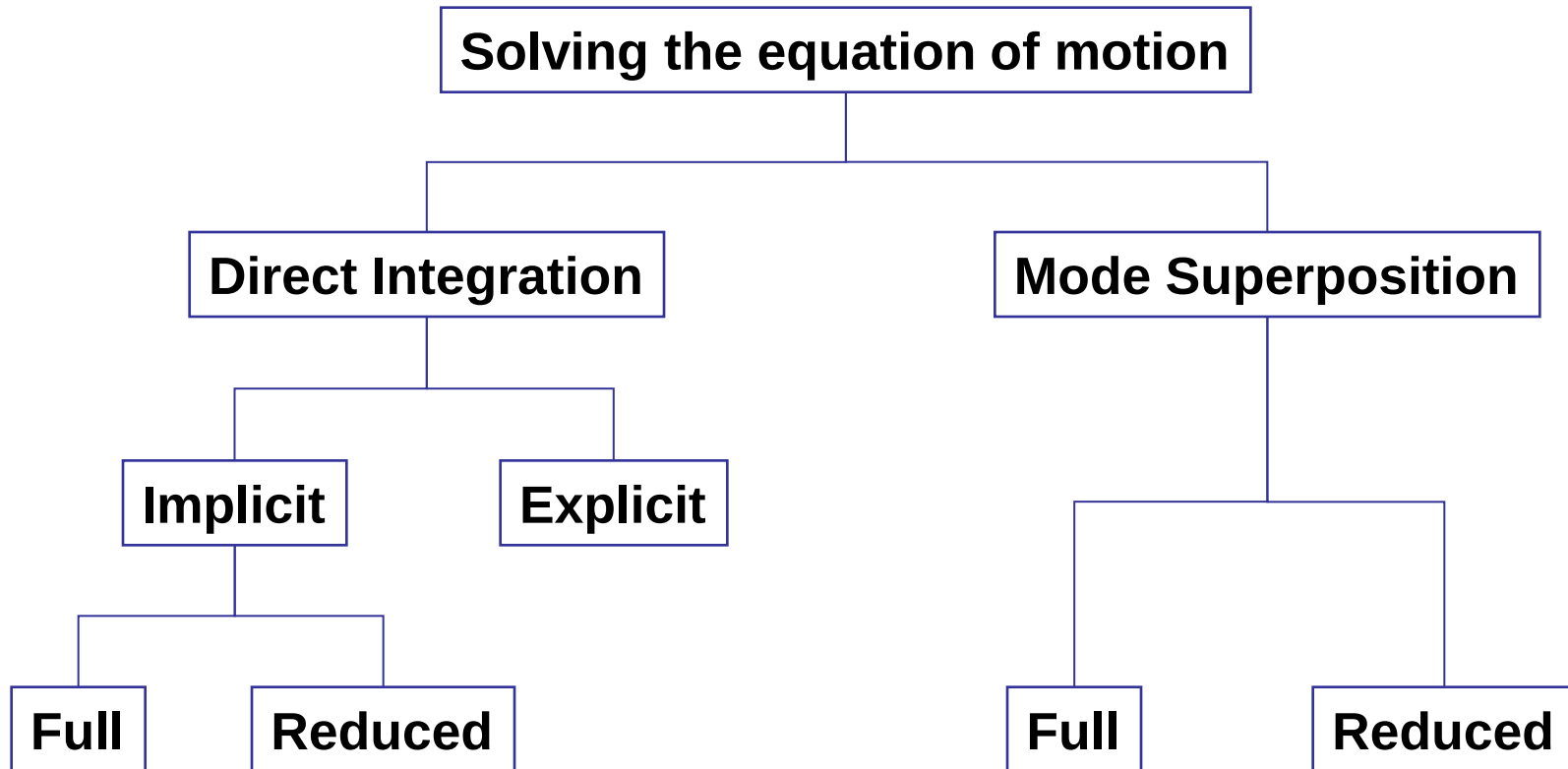
- **Equation of motion for a transient dynamic analysis is the same as the general equation of motion.**

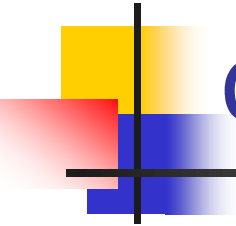
$$[M]\{\ddot{u}\} + [C]\{\dot{u}\} + [K]\{u\} = \{F(t)\}$$

- **This is the most general form of dynamic analysis. Loading may be any arbitrary function of time.**
- **Depending on the method of solution, ANSYS allows all types of nonlinearities to be included in a transient dynamic analysis - large deformation, contact, plasticity, etc.**



Solution Methods





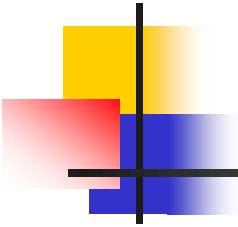
discussion of implicit and explicit

Implicit Method

- **Matrix inversion is required.**
- **Nonlinearities require equilibrium iterations (convergence problems).**
- **Integration time step Δt can be large but may be restricted by convergence issues.**
- **Efficient for most problems except where Δt needs to be very small.**

Explicit Method

- **No matrix inversion.**
- **Can handle nonlinearities easily (no convergence issues).**
- **Integration time step Δt must be small (1e-6 second is typical).**
- **Useful for short duration transients such as wave propagation, shock loading, and highly nonlinear problems such as metal forming.**
- **ANSYS-LS/DYNA uses this method.**



Implicit method

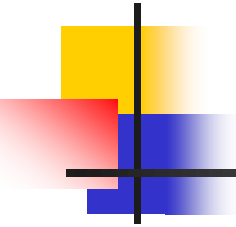
- **ANSYS uses Newmark integration scheme.**

$$[M]\{\ddot{u}_{t+\Delta t}\} + [C]\{\dot{u}_{t+\Delta t}\} + [K]\{u_{t+\Delta t}\} = \{F(t)\}$$

$$u_{t+\Delta t} = u_t + \dot{u}_t \Delta t + [(1/2 - \alpha)\ddot{u}_t + \alpha\ddot{u}_{t+\Delta t}]\Delta t^2$$

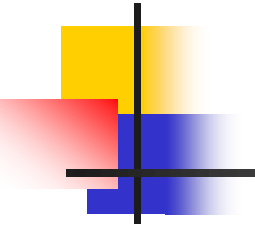
$$\dot{u}_{t+\Delta t} = \dot{u}_t + [(1 - \beta)\ddot{u}_t + \beta\ddot{u}_{t+\Delta t}]\Delta t$$

- **Varying values of α and β causes integration scheme to change (implicit / explicit / average acceleration).**
- **Newmark is an implicit scheme.**
- **LS-DYNA uses explicit scheme.**



Integration Time Step

- The **integration time step** (also **ITS** or Δt) size should be small enough to capture the following:
 - the response frequency
 - the contact frequency (if applicable)
 - wave propagation effects (if applicable)
 - Nonlinear response (plasticity, creep, contact status)



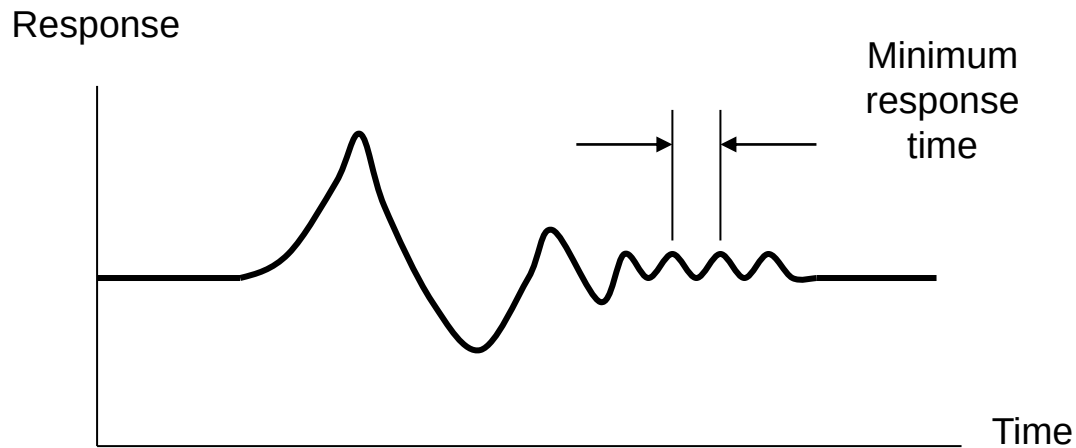
反應頻率(response frequency)

- ❖ 針對一般的暫態動力學問題， **Δt** 概略的選取判斷如下式：

$$\Delta t = \frac{1}{20f}$$

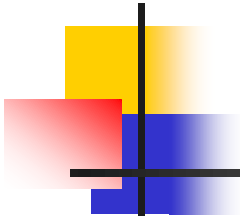
其中 **Δt** 單位為秒， **f** 為結構的反應頻率(**response frequency**)，單位為**Hz**，這是概指分析案例的最高頻率。

- ❖ 至於反應頻率 **f** 值，則必須先經由初步的**ANSYS**暫態動力學分析後才能得到，**ANSYS**在求解時會於**Output Window**列印出 **f** 值，以供參考。接著利用**ANSYS**提供的 **f** 值，再以上式求出適當的 **Δt** ，以此 **Δt** 再重做一次分析。



Δt 要足夠小到能捕捉結構反應的細節，上圖表示結構的反應 (**structural response**)，假設最小的反應周期是 **T** ，則 **Δt** 大致上不要超過 **$T/20$** 。

$$\Delta t \leq \frac{T}{20}$$



接觸頻率(contact frequency)

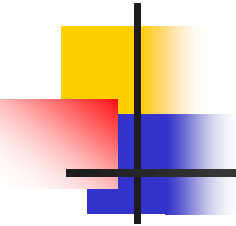
- ❖ 若暫態動力學分析包含了接觸問題，則需要考慮 Δt 是否能精確模擬出兩物體接觸時的動量轉移，若 Δt 不夠小則會造成誤差。對於這類問題的 Δt 概略取法如下式：

$$\Delta t = \frac{1}{30f_c}$$

$$f_c = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

其中 f_c 稱為接觸頻率(**contact frequency**)， k 為間隙剛度(**gap stiffness**)， m 為接觸面的等效質量(**effective mass**)。

- ❖ 不過，上式的 k 和 m 較不易明確定義，而**ANSYS**在求解時會於**Output Window**列印出反應頻率(**response frequency**)，即 f_c 值，以供使用者參考。



應力波傳遞(wave propagation)

❖ 若要求解應力波傳遞問題，其 Δt 概略取法如下式：

$$\Delta t \leq \frac{\Delta x}{3c}$$

$$c = \sqrt{E/\rho}$$

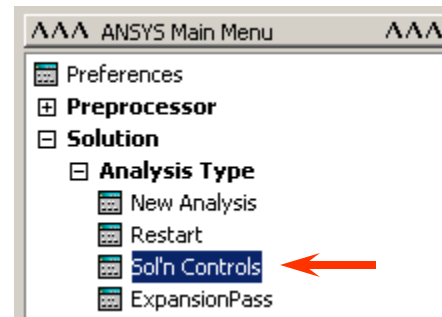
其中 Δx 為元素尺寸，若波傳遞方向的結構長度為 L ，則 Δx 須小於 $L/20$ ， c 則為彈性波速(**elastic wave speed**)，而 E 和 ρ 分別為楊氏模數和密度。以上計算須注意單位的相容性。

❖ Δt 的設定需由**ANSYS**使用者給定，而**ANSYS**提供了指令**AUTOTS**和**DELTIM**，可自動調整時間增量。

Choose Analysis Type & Options

Solution options

- Choose large displacement transient or small displacement transient .
 - When in doubt, choose large displacement transient



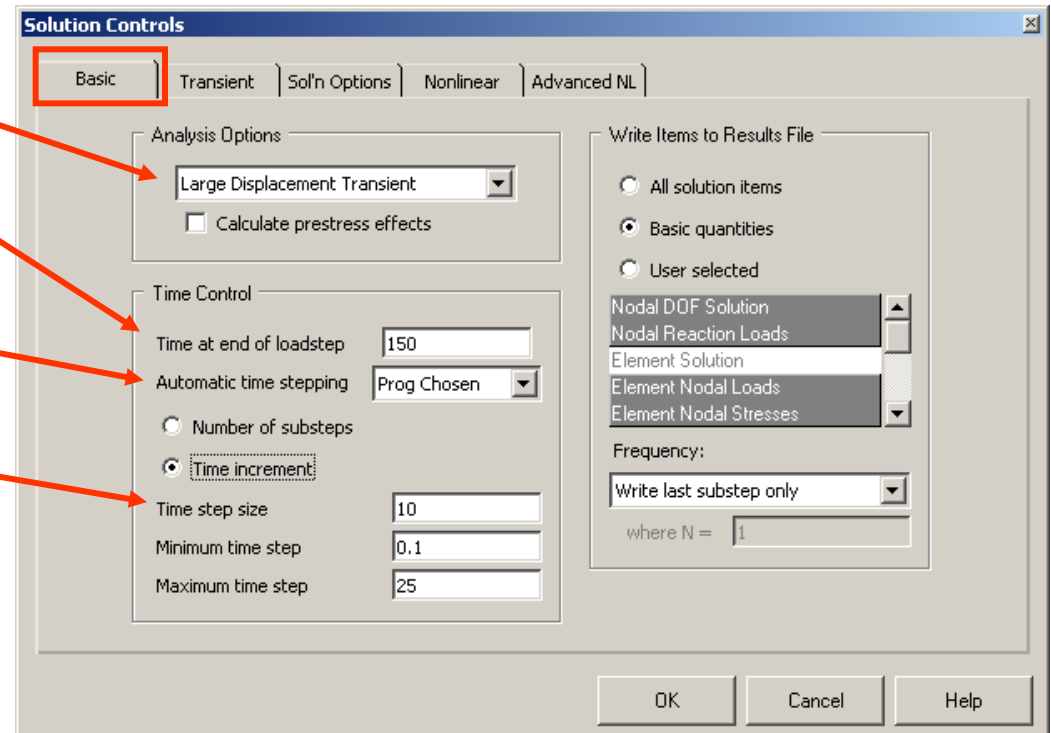
Specify time at end of load step.

Automatic time stepping

- If nonlinearities are present, use the "Program Chosen" option.

Specify initial, min and max values of Δt for this load step.

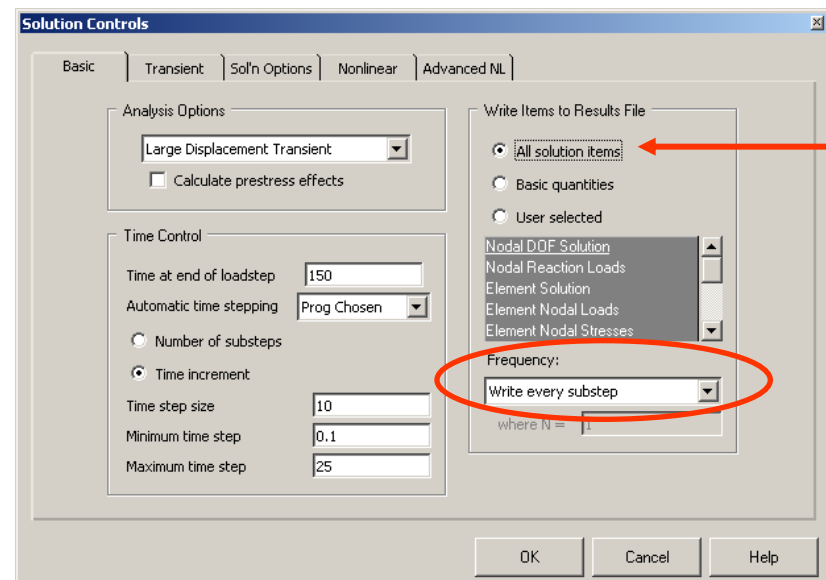
Specify output controls



Choose Analysis Type & Options

Output controls

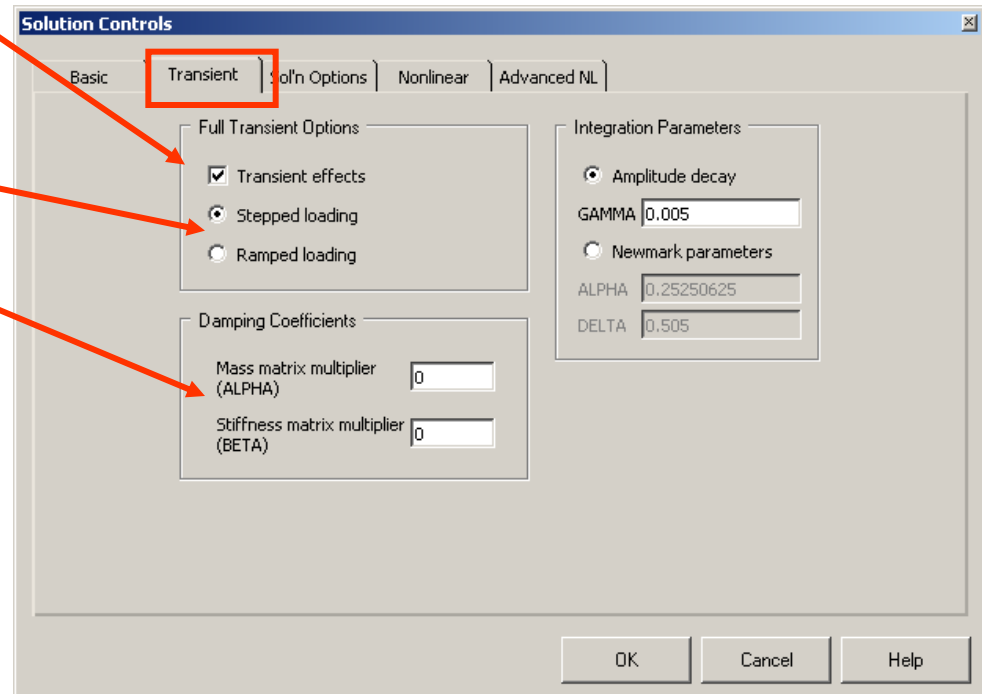
- Used to determine what is written to the results file.
- Use the **OUTRES** command or choose **Solution > Sol'n Control.. > Basic** in the menu
- Typical choice is to write all items at every substep to the results file.
 - Allows smooth plots of results vs. time.
 - Might cause results file to be large.



Turn transient effects on/off

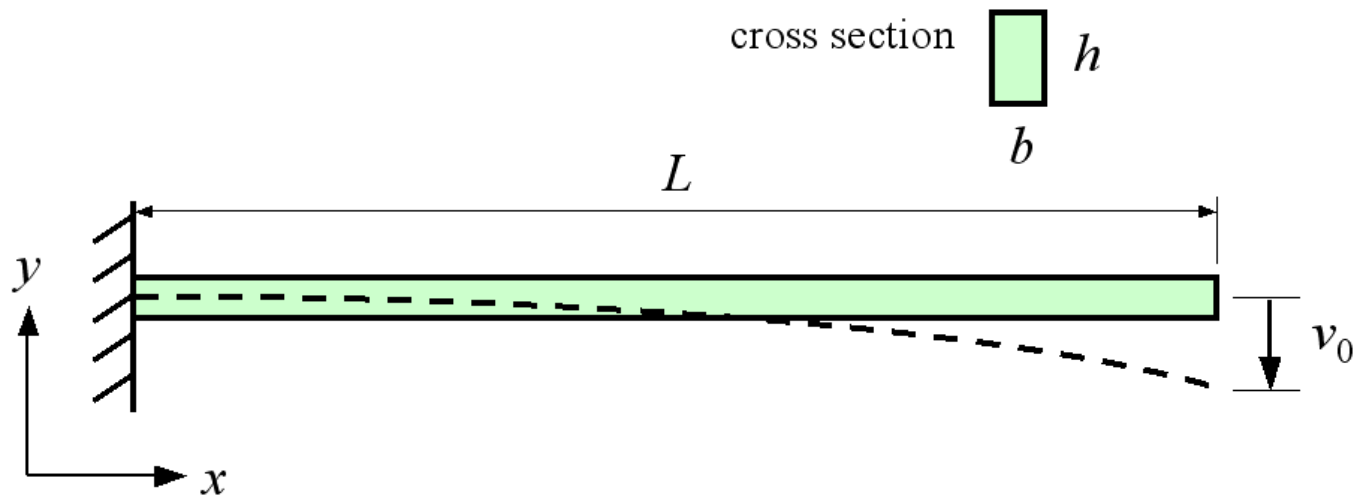
- useful for setting up initial conditions (discussed later)

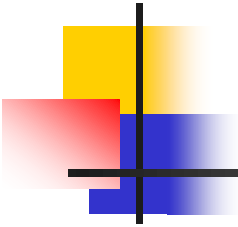
- **Ramp** or **Step** apply load
- **Specify damping** (discussed later)
- **Use default values for time integration parameters**



懸臂樑暫態反應分析

如 下圖 的二維懸臂樑問題，樑之長度 $L = 6 \text{ m}$ ，矩形截面 $b = 200 \text{ mm}$ 和 $h = 400 \text{ mm}$ ，楊氏模數 $E = 12 \text{ GPa}$ ，密度 $\rho = 1500 \text{ kg/m}^3$ ，假設其 Rayleigh 阻尼係數 $\alpha = 0$ 和 $\beta = 0.006$ 。懸臂樑初始條件為施加一負 y 方向位移 $v_0 = 40 \text{ mm}$ 於右端，且其初始速度 $\dot{v}_0 = 0$ 。試求右端的初始拘束釋放後，右端在 2 秒內的暫態位移響應，分析單位系統採用： m 、 N 、 Pa 、 kg 。





Rayleigh damping

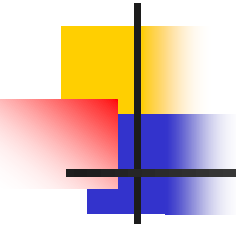
- 又稱為比例阻尼(**proportional damping**)，它是假設整個結構之阻尼模式均相同，如下為結構之運動方程式：

$$[M]\{\ddot{u}\} + [C]\{\dot{u}\} + [K]\{u\} = \{F\}$$

其中阻尼矩陣[**C**]可以下列方式表示：

$$[C] = \alpha [M] + \beta [K]$$

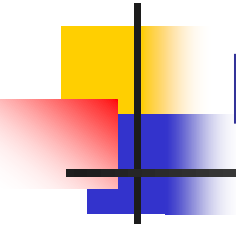
$\alpha[M]$ 和 $\beta[K]$ 的意義是分別用來表現**viscous damping**(結構和周遭流體的摩擦)和**hysteresis damping**(材料本身內部分子間的摩擦)。



Specify BC's & Initial Conditions

Example - “Plucking” the free end of a cantilever beam

- In this case $u_0 \neq 0$ at one end of the beam, and $v_0 = 0$.
- Use the static load step method.
- Load step 1:
 - Transient effects OFF. Use TIMINT,OFF command or
 Solution > Sol'n Control
 Select the “Transient” Tab and unselect Transient effects
 - Small time interval, e.g, 0.001.
 - 2 substeps, stepped loads. (If ramped or with one substep, v_0 will be non-zero.)
 - Apply the desired non-zero displacement at the free end of the beam.
 - SOLVE.
- Load step 2:
 - Transient effects ON.
 - Delete the imposed displacement.
 - Specify ending time and continue with the transient.



Example: Plucking a Cantilever Beam

```
01  /SOLU
02    ANTYPE, TRANS
03    ...
04    TIMINT, OFF  ! Transient effects off
05    TIME, 0.001  ! Small time interval
06    D, ...       ! Apply displacement at desired nodes
07    KBC, 1       ! Stepped loads
08    NSUBST, 2    ! To avoid non-zero velocity
09    SOLVE
10
11    TIMINT, ON    ! Transient effects on
12    TIME, ...     ! Actual time at end of load
13    DDELE, ...   ! Delete the applied displacement
14    SOLVE
```

如果跳出/SOLU，再進入/SOLU，靜力分析的結果會因重新進入/SOLU後被清除掉

Output Window 印出的反應頻率

```
ANSYS 7.0 Output Window

*** LOAD STEP      2    SUBSTEP   131  COMPLETED.    CUM ITER =   133
*** TIME =    1.67169      TIME INC = 0.128846E-01  OLD TRIANG MATRIX
*** RESPONSE FREQ =  5.072      PERIOD= 0.1971      PTS/CYC = 15.
*** AUTO STEP TIME:  NEXT TIME INC = 0.12885E-01  UNCHANGED

*** LOAD STEP      2    SUBSTEP   132  COMPLETED.    CUM ITER =   134
*** TIME =    1.68457      TIME INC = 0.128846E-01  OLD TRIANG MATRIX
*** RESPONSE FREQ =  5.072      PERIOD= 0.1971      PTS/CYC = 15.
*** AUTO STEP TIME:  NEXT TIME INC = 0.12885E-01  UNCHANGED

*** LOAD STEP      2    SUBSTEP   133  COMPLETED.    CUM ITER =   135
*** TIME =    1.69746      TIME INC = 0.128846E-01  OLD TRIANG MATRIX
*** RESPONSE FREQ =  5.072      PERIOD= 0.1971      PTS/CYC = 15.
*** AUTO STEP TIME:  NEXT TIME INC = 0.12885E-01  UNCHANGED

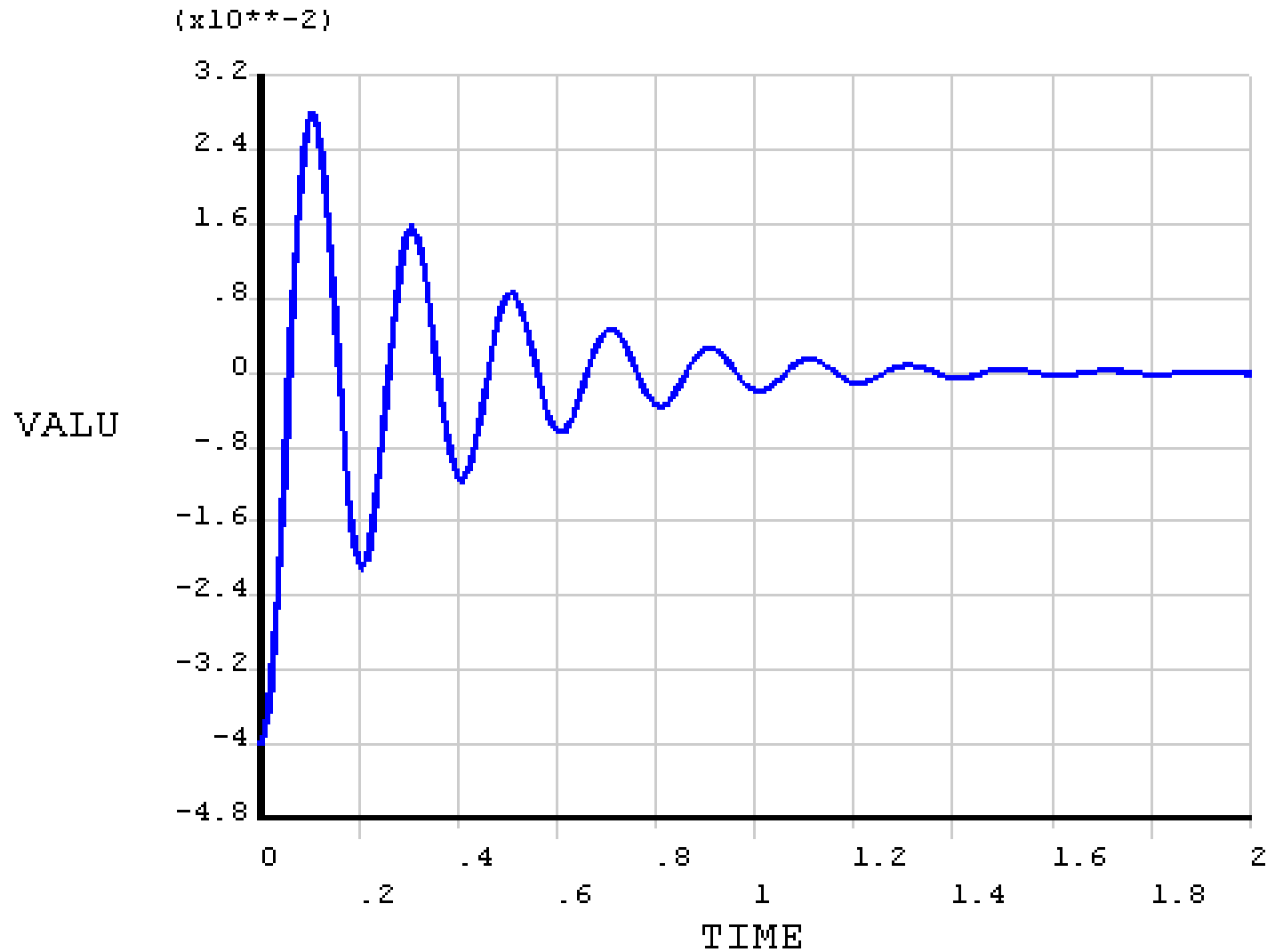
*** LOAD STEP      2    SUBSTEP   134  COMPLETED.    CUM ITER =   136
*** TIME =    1.71034      TIME INC = 0.128846E-01  OLD TRIANG MATRIX
*** RESPONSE FREQ =  5.072      PERIOD= 0.1971      PTS/CYC = 15.
*** AUTO STEP TIME:  NEXT TIME INC = 0.12885E-01  UNCHANGED

*** LOAD STEP      2    SUBSTEP   135  COMPLETED.    CUM ITER =   137
*** TIME =    1.72323      TIME INC = 0.128846E-01  OLD TRIANG MATRIX
*** RESPONSE FREQ =  5.072      PERIOD= 0.1971      PTS/CYC = 15.
*** AUTO STEP TIME:  NEXT TIME INC = 0.12885E-01  UNCHANGED

*** LOAD STEP      2    SUBSTEP   136  COMPLETED.    CUM ITER =   138
*** TIME =    1.73611      TIME INC = 0.128846E-01  OLD TRIANG MATRIX
```

反應頻率為5.072Hz，由此算出 Δt 為 $1/(20*5.072) \approx 0.0099$ 秒，約為0.01秒，因此DELTIM指令中的起始副增量可設定為0.01秒。

樑右端在2秒內的位移變化



位移單位：m
時間單位：sec